



Deep Learning in pricing: The role of information

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- 1. Kullback-Leibler Divergence**
- 2. Mutual Information**

Kullback-Leibler Divergence



$$D_{KL}[p||q] = \int_{-\infty}^{\infty} p(x) \text{Log} \frac{p(x)}{q(x)} dx$$

$$D_{KL}[p||p] \geq 0 \quad \text{et} \quad D_{KL}(p||p) = 0$$

But it is not a distance but triangular inégalité does not work

Is not symmetric

let $q(x)$ a prob. distribution defined on E et $F \subset E$

Let $q_F(x)$ a prob. distribution restricted to F (conditionned by F)

$$\text{Then } D_{KL}[q||q_F] = \text{Log} \frac{1}{\text{Prob}[F]}$$

- This is the supplementary information: conditionned by F (in bits)
- This is the Renyi divergence for $\alpha = 1$, $D_\alpha[p||q] = \frac{1}{1-\alpha} \int_{-\infty}^{\infty} p(x)^\alpha q(x)^{1-\alpha} dx$ Hellinger

The topology generated by D_{KL} and D_α are not the same : $D_{1/2} \rightarrow \text{Distance}$ $\downarrow$

$$\lim_{\theta_1 \rightarrow \theta} \frac{1}{(\theta_1 - \theta)^2} D_\alpha[p_{\theta_1}||p_\theta] = \frac{\alpha}{2} J(\theta) \leftarrow \text{Fisher Information}$$

KL-Divergence and Crossed Entropy..



Choosing
The right tool



Claude Shannon

$$D_{KL}[p||q] = \int p(x) \log p(x) dx - \int p(x) \log q(x) dx$$

Measure a spread

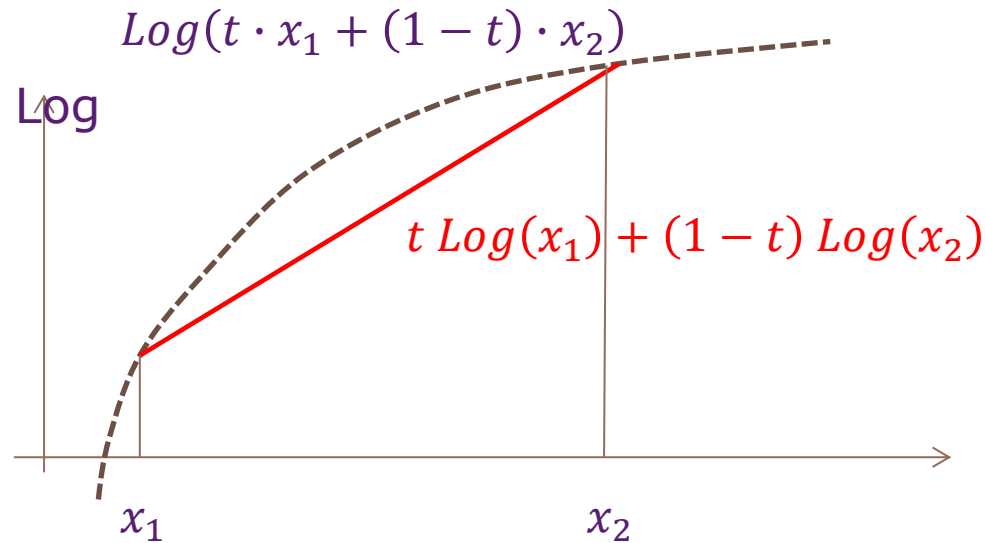
$$D_{KL}[p||q] = H(p) - H(p, q)$$

Entropy

Crossed Entropy

- An Efficient criteria
- Classification problems
- Far Gradient still active

Jensen Inequality



2 points

N points > 2

Jensen :

$$F \text{ concave} \Rightarrow f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$$

Inférence variationnelle (mécanique)



$$\text{Log } p(x) - D_{KL}[q(z)||p(z|x)] = (\mathbb{E}_q[\text{Log } p(z,x)] - \mathbb{E}_q[\text{Log } q(z)])$$

Or $p(z, x) = p(x|z) p(z)$

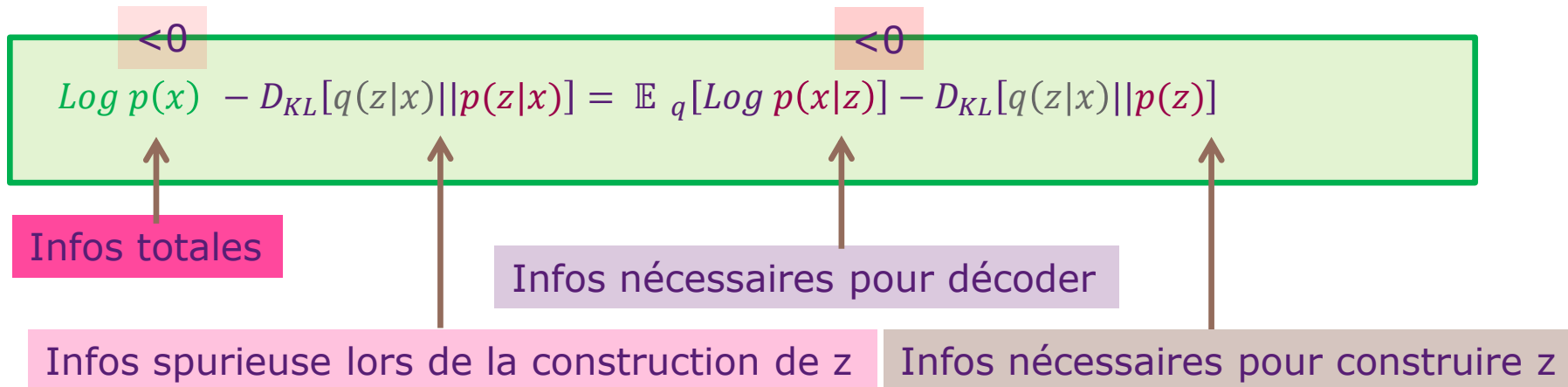
$$\text{Log } p(x) - D_{KL}[q(z)||p(z|x)] = (\mathbb{E}_q[\text{Log } p(x|z) + \text{Log } p(z)] - \mathbb{E}_q[\text{Log } q(z)])$$

Or $D_{KL}[q(z)||p(z)] = \mathbb{E}_q[\text{Log } p(z)] - \mathbb{E}_q[\text{Log } q(z)]$

$$\text{Log } p(x) - D_{KL}[q(z)||p(z|x)] = \mathbb{E}_q[\text{Log } p(x|z)] - D_{KL}[q(z)||p(z)]$$



On peut faire dépendre q de x



Variational Inference (Boundary)



X observed
Z hidden

Z is not observed

$$\text{Log } p(x) = \text{Log} \left[\int_z p(x, z) dz \right]$$

q(z) arbitrary

$$= \text{Log} \left[\int_z p(x, z) \frac{q(z)}{q(z)} dz \right]$$

$$= \text{Log} \left[\mathbb{E}_q \left[\frac{p(x, z)}{q(z)} \right] \right]$$

Entropy of z

Jensen

$$\geq \mathbb{E}_q [\text{Log}[p(x, z)]] - \mathbb{E}_q [\text{Log}[q(z)]] \equiv \text{ELBO}(p, q)$$

Then maximising expectation of the low bound gives a low bound on the likelihood which is a good approximation

Relationship with KL divergence



We want to approximate $p(z|x) = \frac{p(z,x)}{p(x)}$ by $q(z)$

Definition

$$\begin{aligned} D_{KL}[q(z)||p(z|x)] &\stackrel{\text{Definition}}{=} \mathbb{E}_q \left[\text{Log} \frac{q(z)}{p(z|x)} \right] \\ &= \mathbb{E}_q [\text{Log} q(z)] - \mathbb{E}_q [\text{Log} p(z|x)] \\ &= \mathbb{E}_q [\text{Log} q(z)] - \mathbb{E}_q [\text{Log} p(z,x)] + \text{Log} p(x) \end{aligned}$$

Doesn't depend on q

$$\text{Equation K} \quad = -(\mathbb{E}_q [\text{Log} p(z,x)] - \mathbb{E}_q [\text{Log} q(z)]) + \text{Log} p(x)$$

Only varying $q(z)$
Then Minimizing ELBO $\Leftrightarrow$ Maximizing KL divergence

Mutual Information Properties



Definition

$$I(X; Y) = D_{KL}[p(x, y) || p(x)p(y)]$$

The information improvement compared to an independent case

Data processing Inequality (DPI)

For any markov chains $X \rightarrow Y \rightarrow Z$

$$I(X; Y) \geq I(X; Z)$$

There is less information on X in Z than in Y

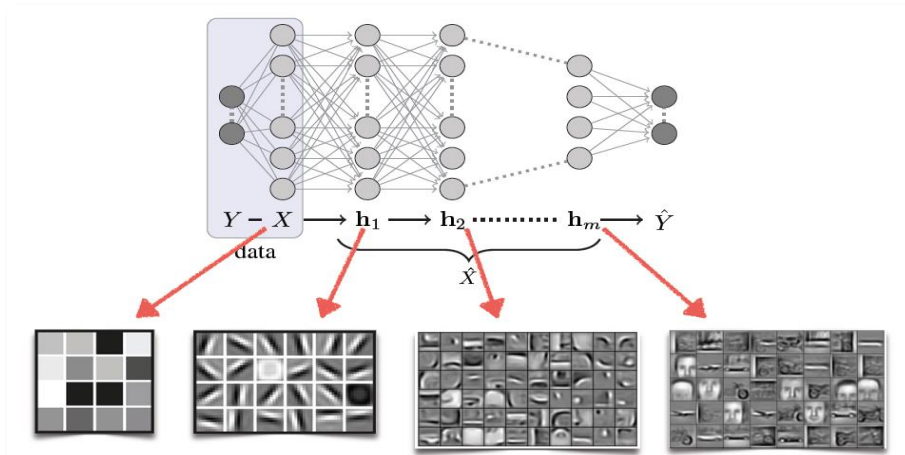
Invariance by reparametrisation

$$I(X; Y) = I(\varphi(X); \psi(Y)) \text{ For any invertible } \varphi \text{ and } \psi$$

First interpretations of layers



What do the DNN Layers represent?



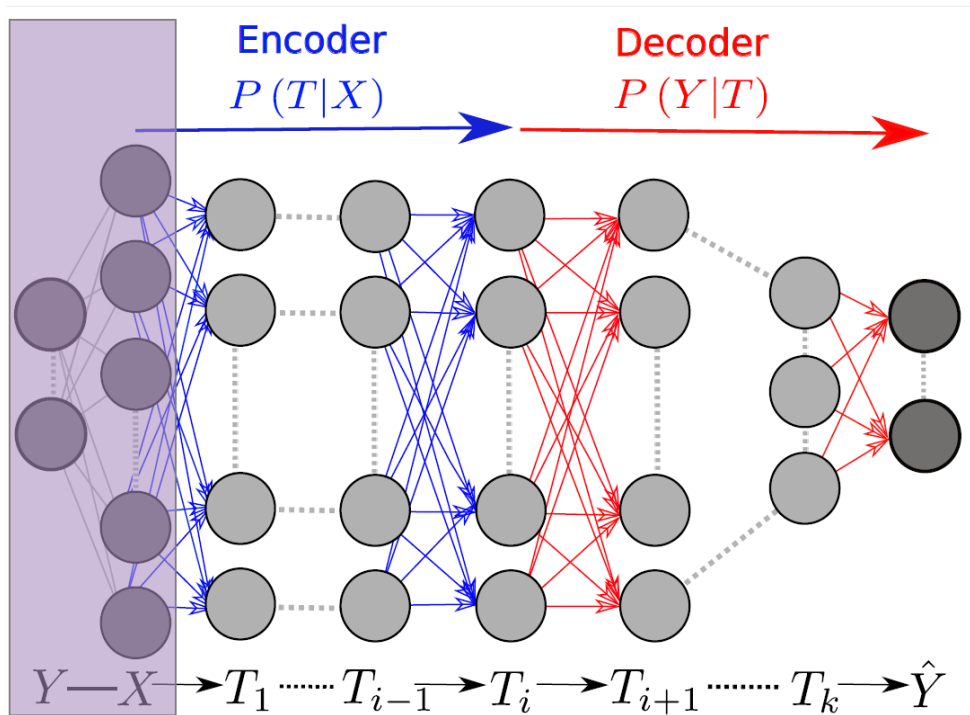
Data Processing Inequalities:

$$H(X) \square I(X; h_i) \square I(X; h_{i+1}) \square I(X; h_{i+2}) \square \dots$$

$$I(X; Y) \square I(h_i; Y) \square I(h_{i+1}; Y) \square I(h_{i+2}; Y) \square \dots$$

- A Markov chain of topologically distinct [soft] partitions of the input variable X .
- Successive Refinement of Relevant Information
- Individual neurons can be easily “scrambled” within each layer

Each layer is characterized by its Encoder & Decoder Information



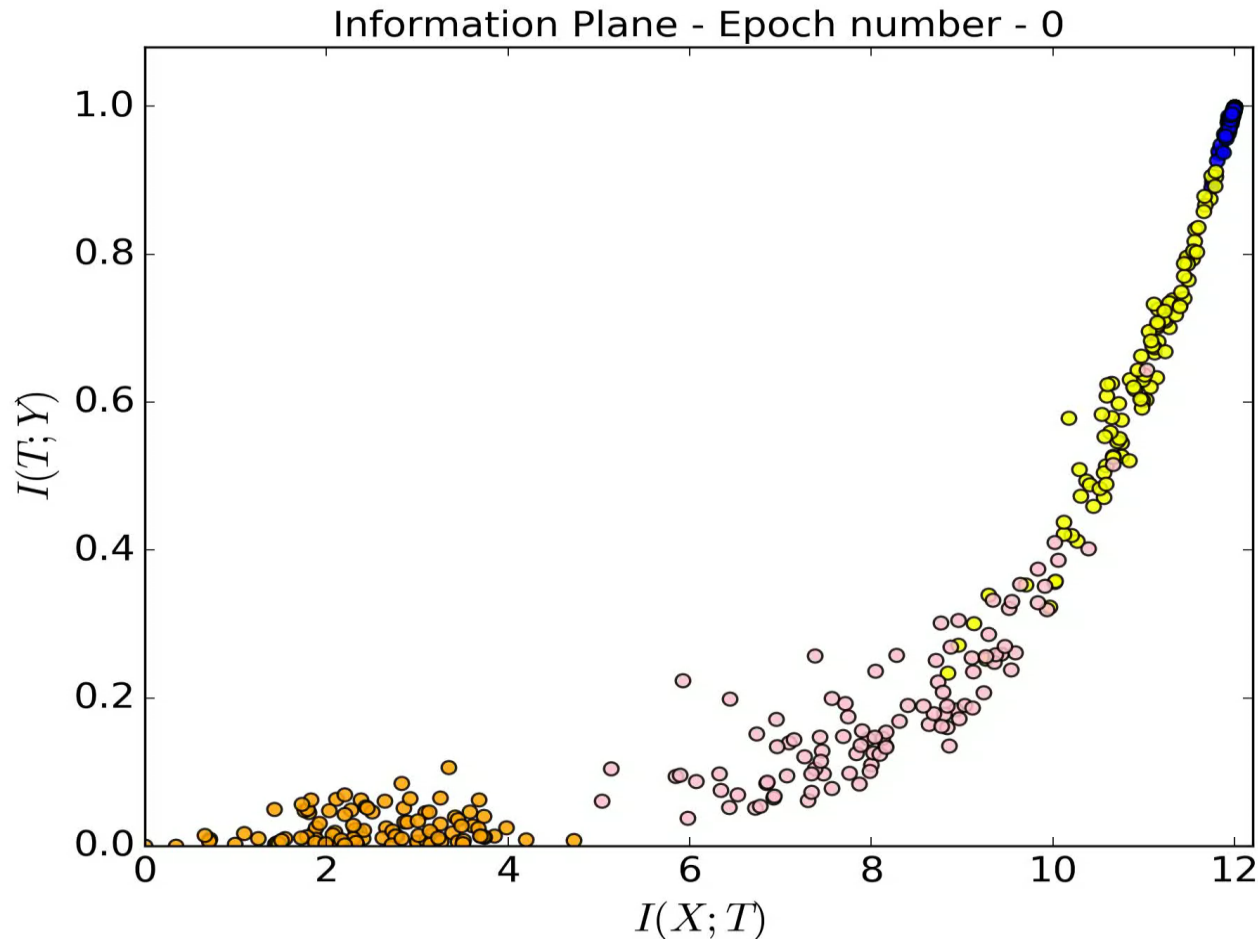
Theorem

(Information Plane):

For large typical $\mathbf{X}$, the sample complexity of a DNN is completely determined by the encoder mutual information, $\mathbf{I}(\mathbf{X};\mathbf{T})$, of the last hidden layer; the accuracy (generalization error) is determined by the decoder information, $\mathbf{I}(\mathbf{T};\mathbf{Y})$, of the last hidden layer.

The complexity of the problem shifts from the decoder to the encoder, across the layers...

100 DNN Layers in Info-Plane without averaging



- Is this the general picture?
- Why do the MI values concentrate?
- What do they mean?
- What governs their dynamics?

The Information Bottleneck optimality bound. (Tishby, Pereira, Bialek, 1999)



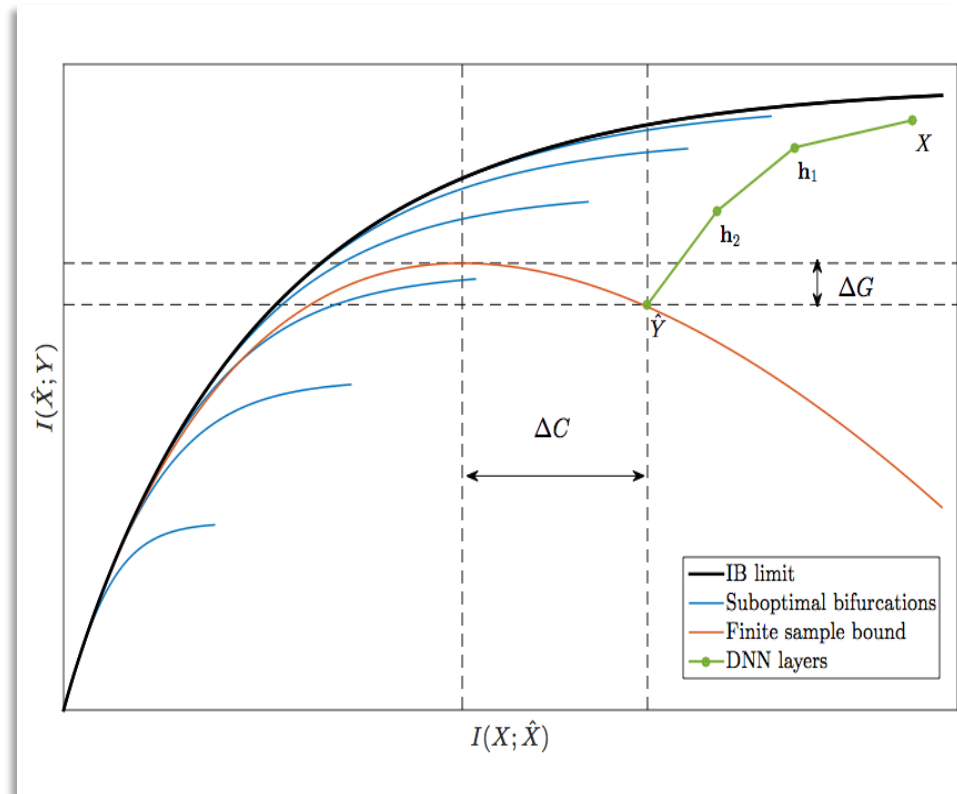
The IB bound optimality equations:

$$\min_{p(\hat{x}|x): Y \rightarrow X \rightarrow \hat{X}} I(\hat{X}; X) - \beta I(\hat{X}; Y), \quad \beta > 0$$

$$\left\{ \begin{array}{l} p(x|\hat{x}) = \frac{p(x)}{Z(x, \beta)} \exp(-\beta D[p(y|x) \| p(y|\hat{x})]) \\ Z(x, \beta) = \sum_x p(\hat{x}) \exp(-\beta D[p(y|x) \| p(y|\hat{x})]) \\ p(\hat{x}) = \sum_x p(\hat{x}|x) p(x) \\ p(y|\hat{x}) = \sum_x p(y|x) p(x|\hat{x}) \end{array} \right.$$

Solved by Arimoto-Blahut like iterations,

but **with possibly sub-optimal solutions, bifurcations (!)**,



Rethinking Learning Theory...



... but we need to guarantee the label homogeneity of the ϵ -partition with finite samples. Without additional structural information on the inputs (stability, robustness, topology), we must use the stochasticity of the rule and the IB distortion measure:

The ϵ -partition, T_ϵ , is with the empirical distortion

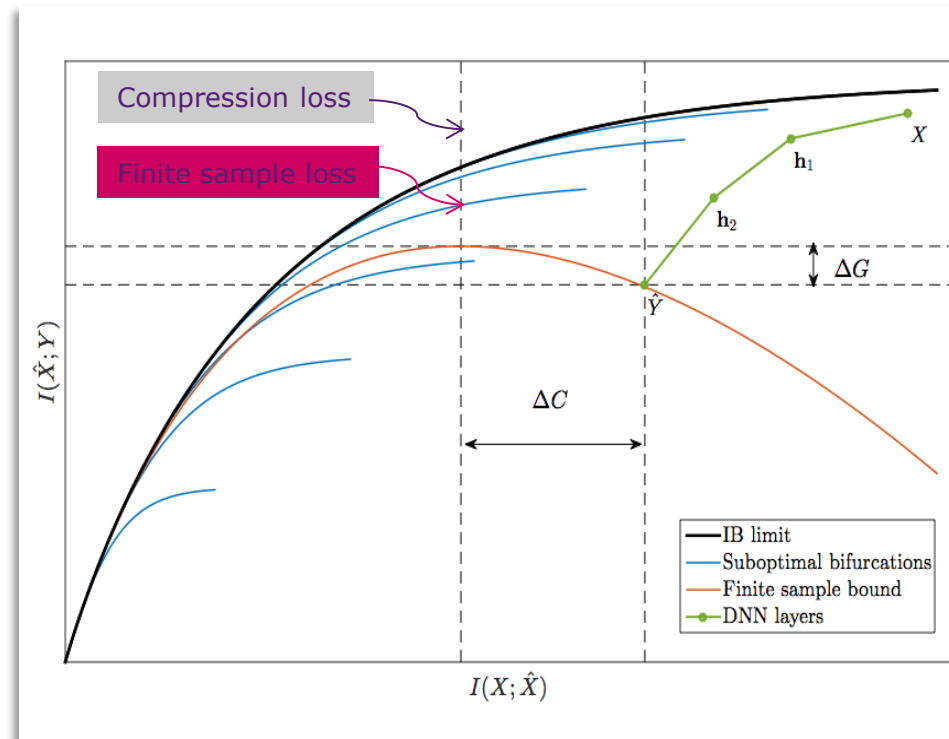
$$d_{IB}(x, t) = D[p_{emp}(y|x) \| p(y|t)]$$

$$\text{as } \langle d_{IB} \rangle_{emp} = I(X; Y) - \hat{I}_{emp}(T; Y)$$

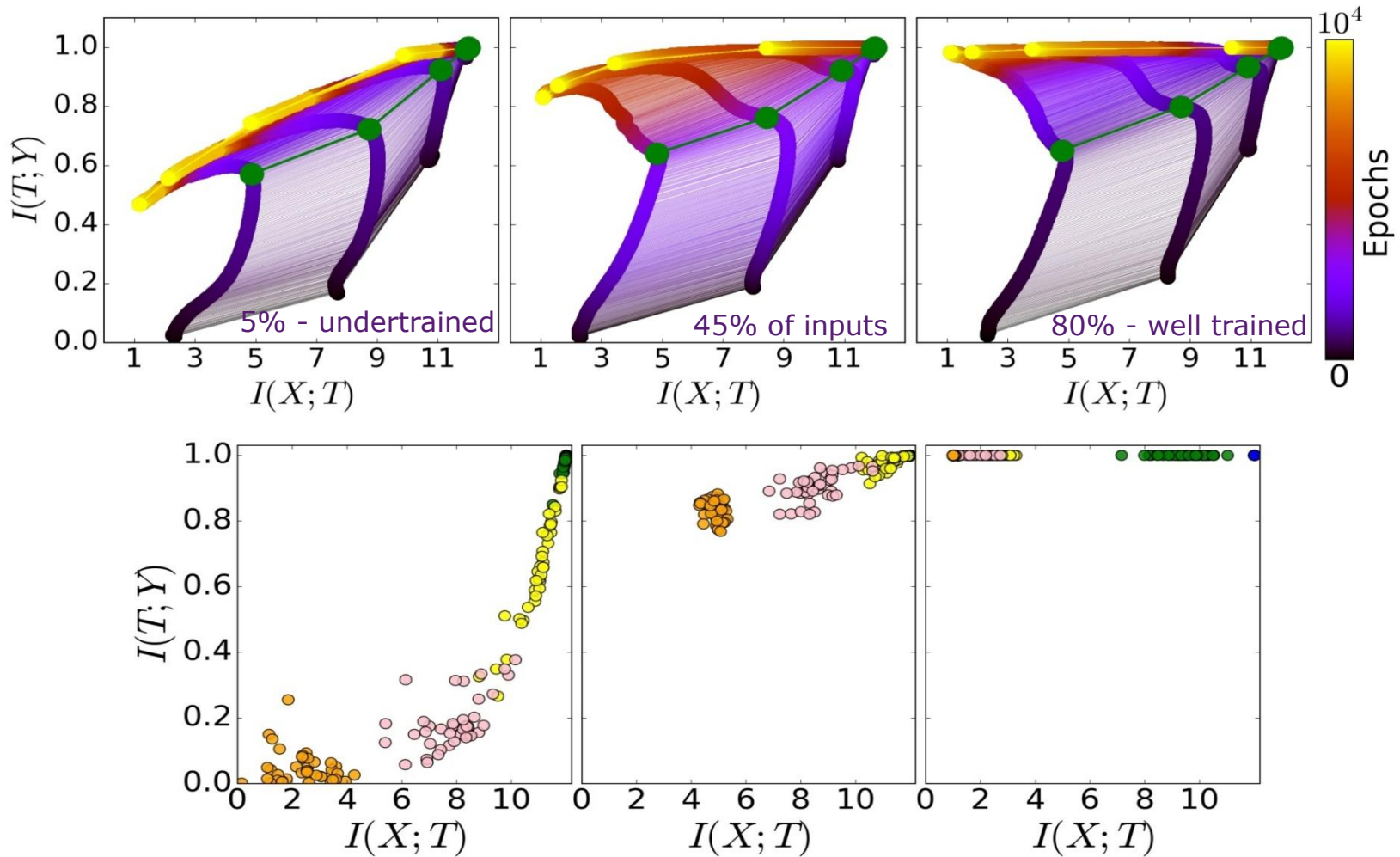
with a finite sample there is another information loss:

$$I(T; Y) \leq \hat{I}_{emp}(T; Y) + O\left(\sqrt{\frac{2^{I(T; X)} |Y|}{m}}\right),$$

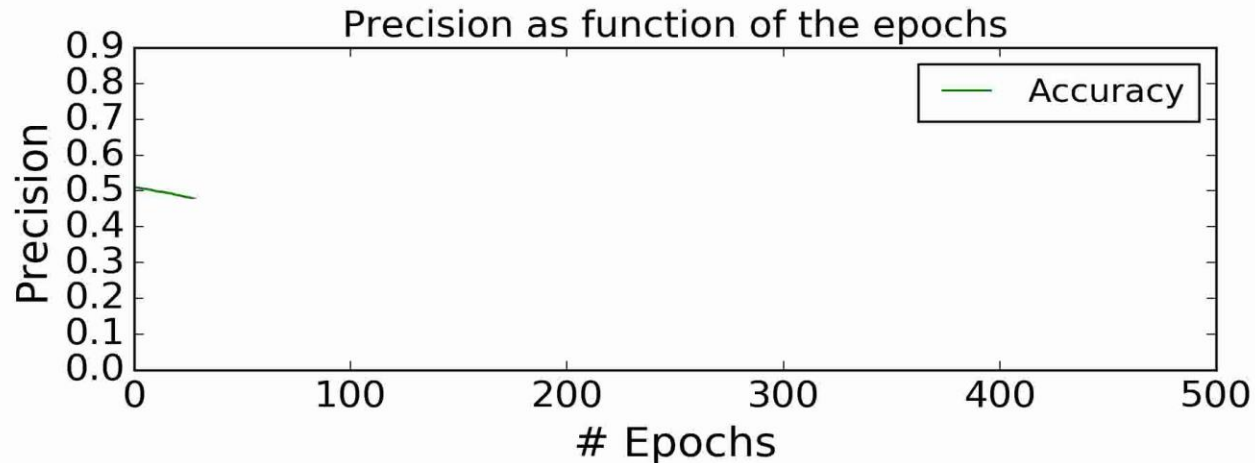
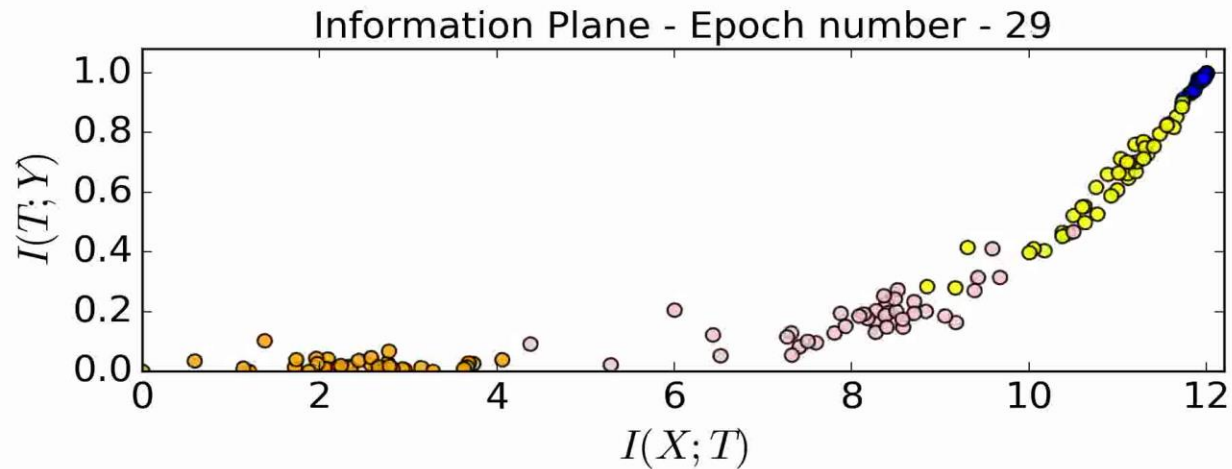
both should remain small for good generalization!



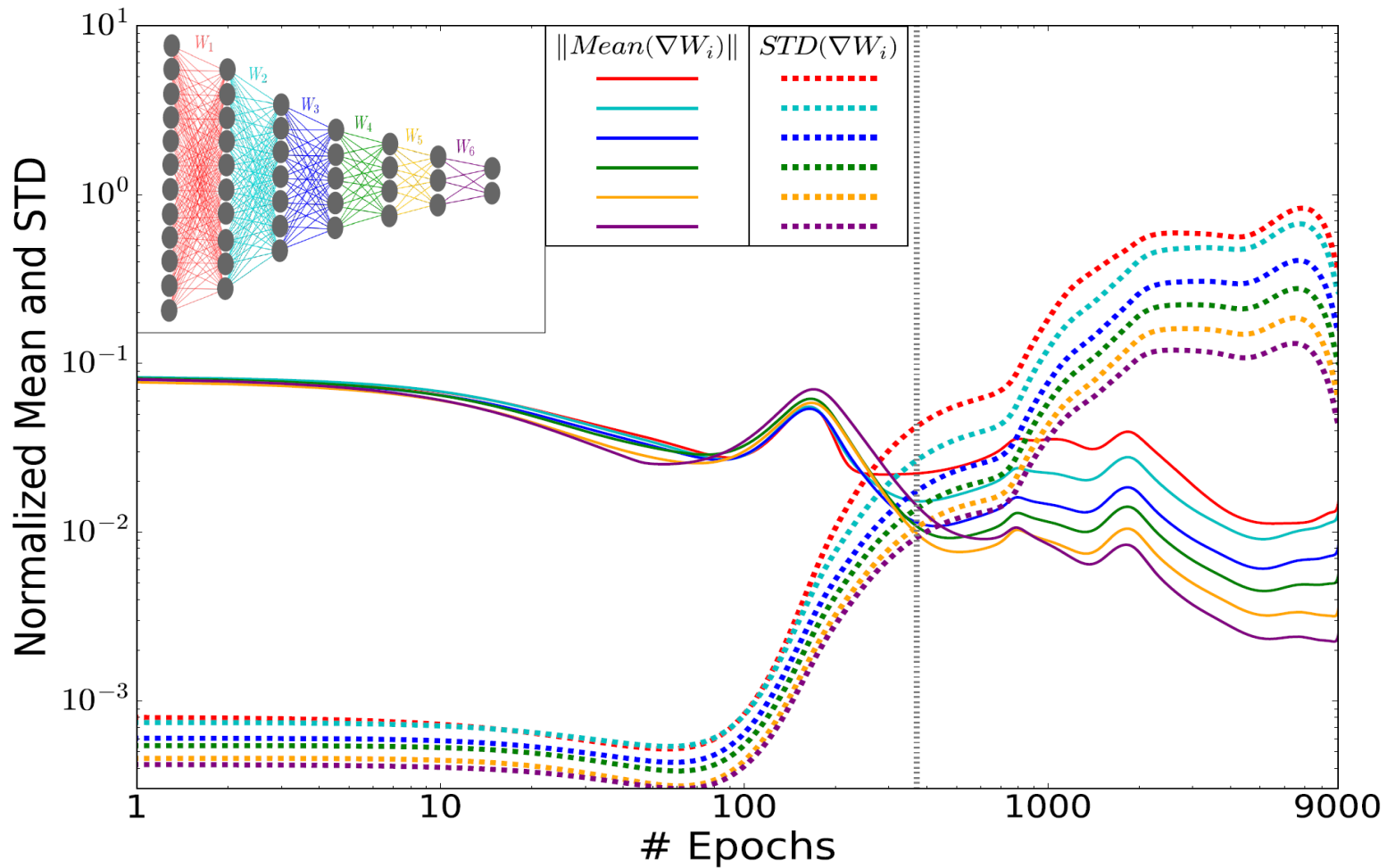
The phase transition comes from the data



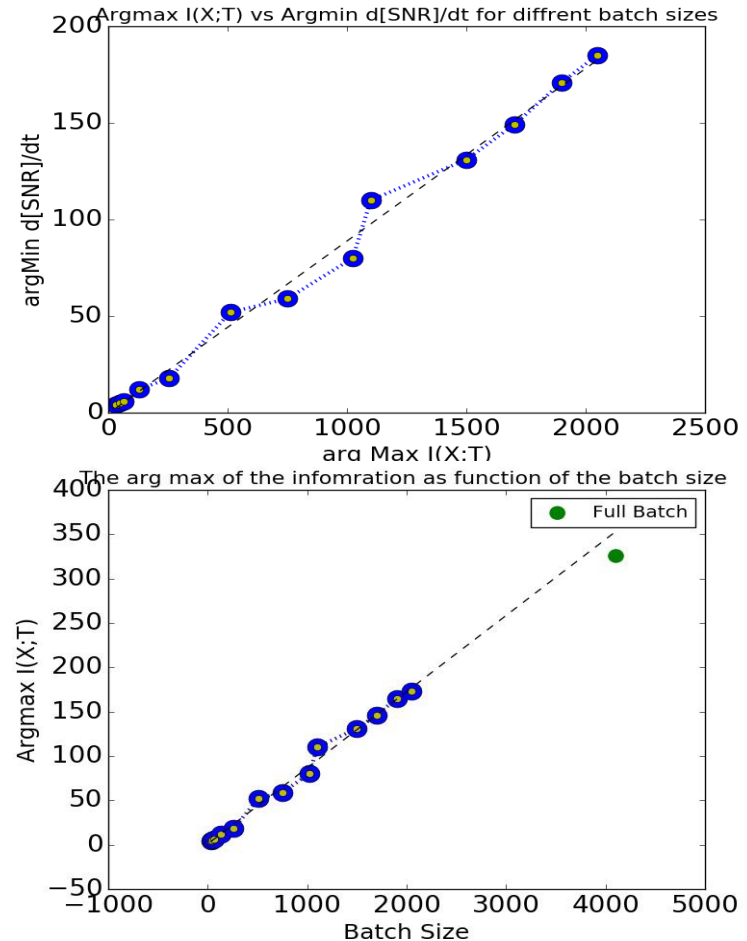
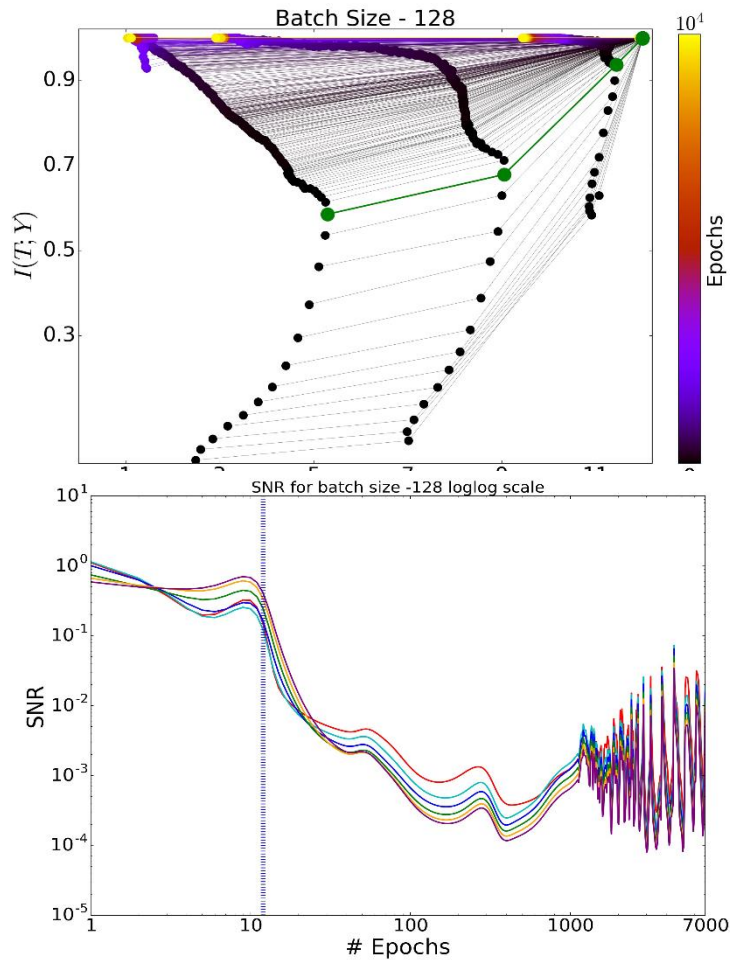
Layers paths with training/generalization error



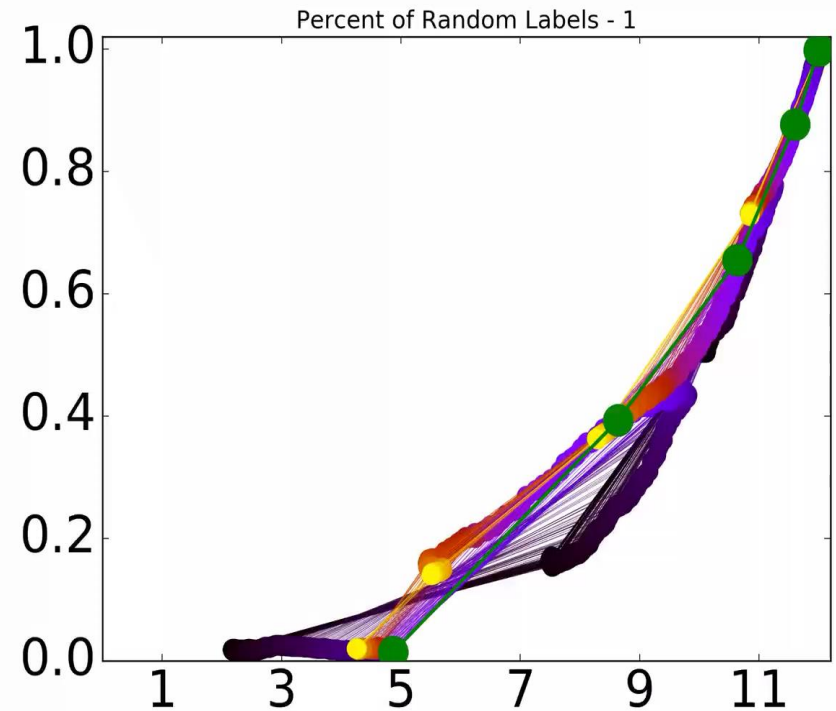
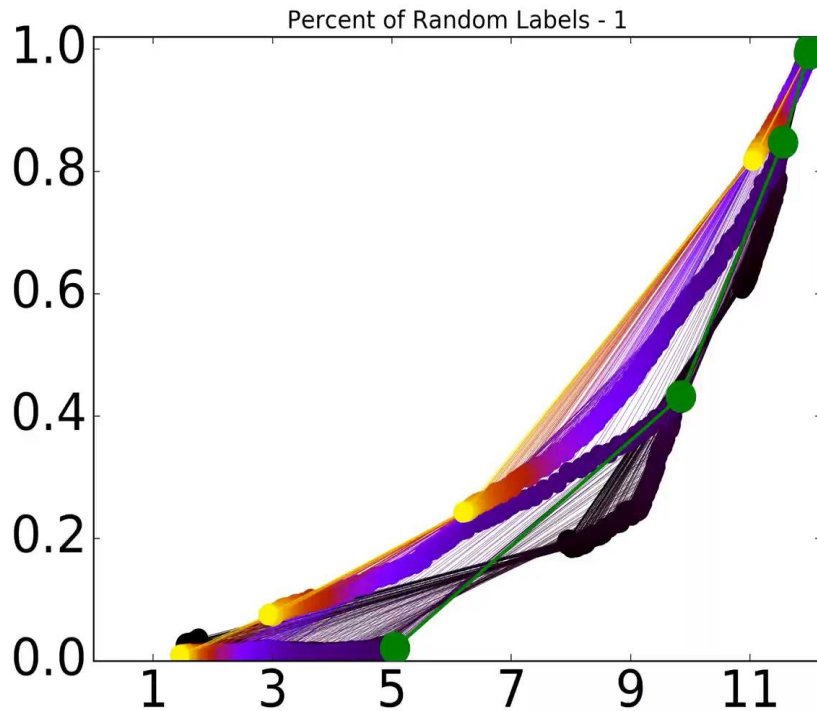
Regim change for the gradients



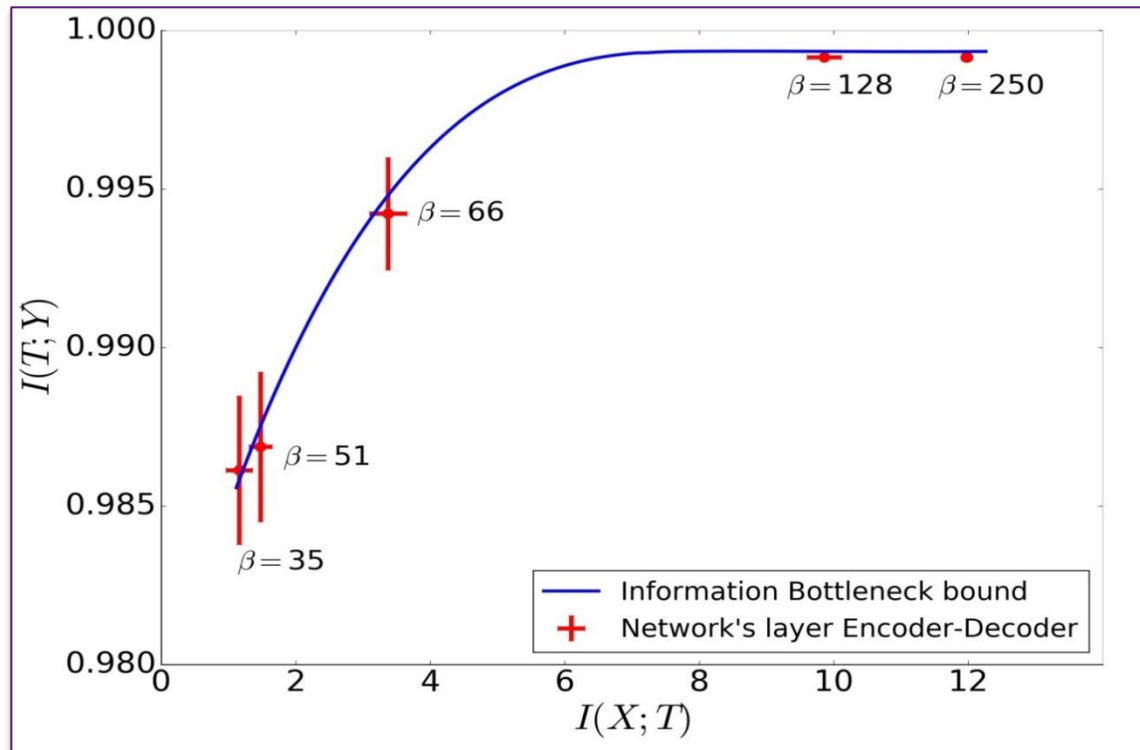
The role of the batch size



Info-plane with random labels



Convergence toward the Btn curve



- Layers of optimal DNN converge to [a successively refineable approximation of] the optimal finite-sample IB limit information-curve
- Layers must be in “different topological phases” of the IB solutions
- The DNN encoder & decoder for each layer satisfy the IB self-consistent equations

Compression through Stochastic Relaxation



Noisy relaxation (SGD) on training error (with m examples):

$$\frac{\partial W_k}{\partial t} = -\nabla E(W_k | X^{(m)}) + \beta_k^{-1} \xi(t), \quad \text{layer } k, \quad \xi \sim N(0,1), \quad \beta_k - \text{decoder } k\text{-layer noise}$$

$$\Rightarrow \text{Maximum Entropy: } P_{Gibbs}(W_k | X^{(m)}) \propto \exp(-\beta_k E(W_k | X^{(m)}))$$

with additive [cross-entropy] training error and i.i.d. samples, using Bayes rule with the quenched W :

$$P_{Gibbs}(X | W_k) = P_{Gibbs}(X | T_k) \propto \exp(-\tilde{\beta}_k D_{KL}[p(Y | X) \| p(Y | T)])$$

This is precisely the IB optimal encoder with $\tilde{\beta}_k$ – the encoder k -layer noise

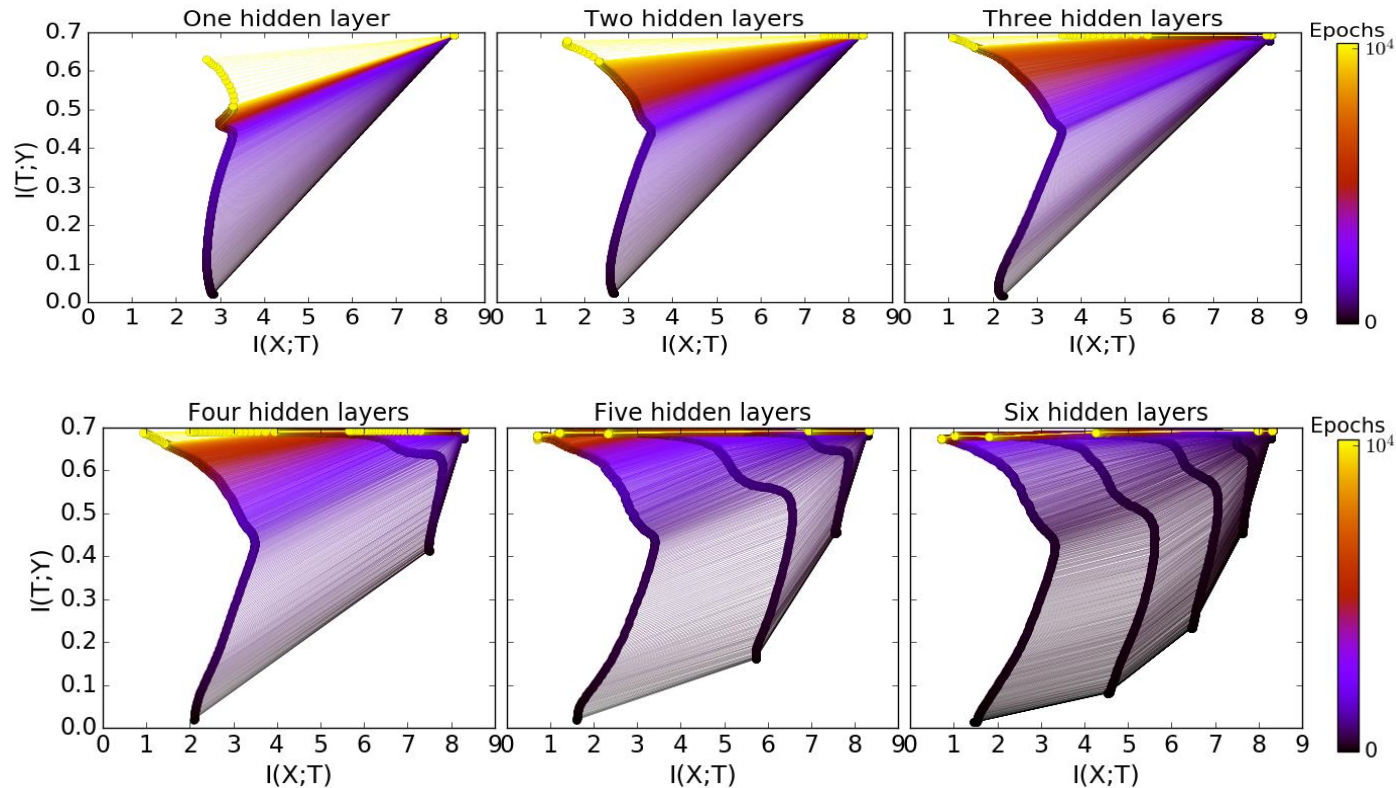
Since $I(X; T_k) = H(X) - H(X | T_k)$, Max Entropy of the weights $\Rightarrow$ Min $I(X; T_k)$

thus SGD converges, layer by layer, to a maximally compressed representation,

which is a SUCCESSIVELY REFINEABLE APPROXIMATION of the optimal IB bound!

Local weights Gibbs and optimal IB representations

The benefit of the hidden layers



- More layers take much FEWER training epochs for good generalization.
- The optimization time depend super-linearly (exponentially?) on the compressed information, ΔI_x , for each layer.

Relaxation times and the benefit of the hidden layers



Noisy relaxation (SGD): $\frac{dW_k}{dt} = -\nabla E(W_k) + \beta_k^{-1} \xi(t)$, layer k , $\xi \sim N(0,1)$

□ Maximum Entropy (via Fokker-Planck): $P_{Gibbs}(W_k) \propto \exp(-\beta_k E(W_k))$,

Relaxation time for non-strongly convex error: $\Delta t_k \propto \exp(\Delta S_k)$

Denote the layer compression be: $\Delta S_k = I(X; T_k) - I(X; T_{k-1})$

Since $\exp\left(\sum_k \Delta S_k\right) \propto \prod_k \exp(\Delta S_k) > \max_k \exp(\Delta S_k) \propto$

Exponential boost in the relaxation time with K layers!

Equilibration of Information Flow through the layers



The Information Capacity between two layers is bounded by the Gaussian capacity:

$$C_G(W_k) \simeq \frac{1}{2} \log \left(1 + \frac{P_k}{N_k} \right) = \frac{1}{2} \log (1 + SNR_k)$$

The stochastic relaxation decreases the SNR of the irrelevant channels

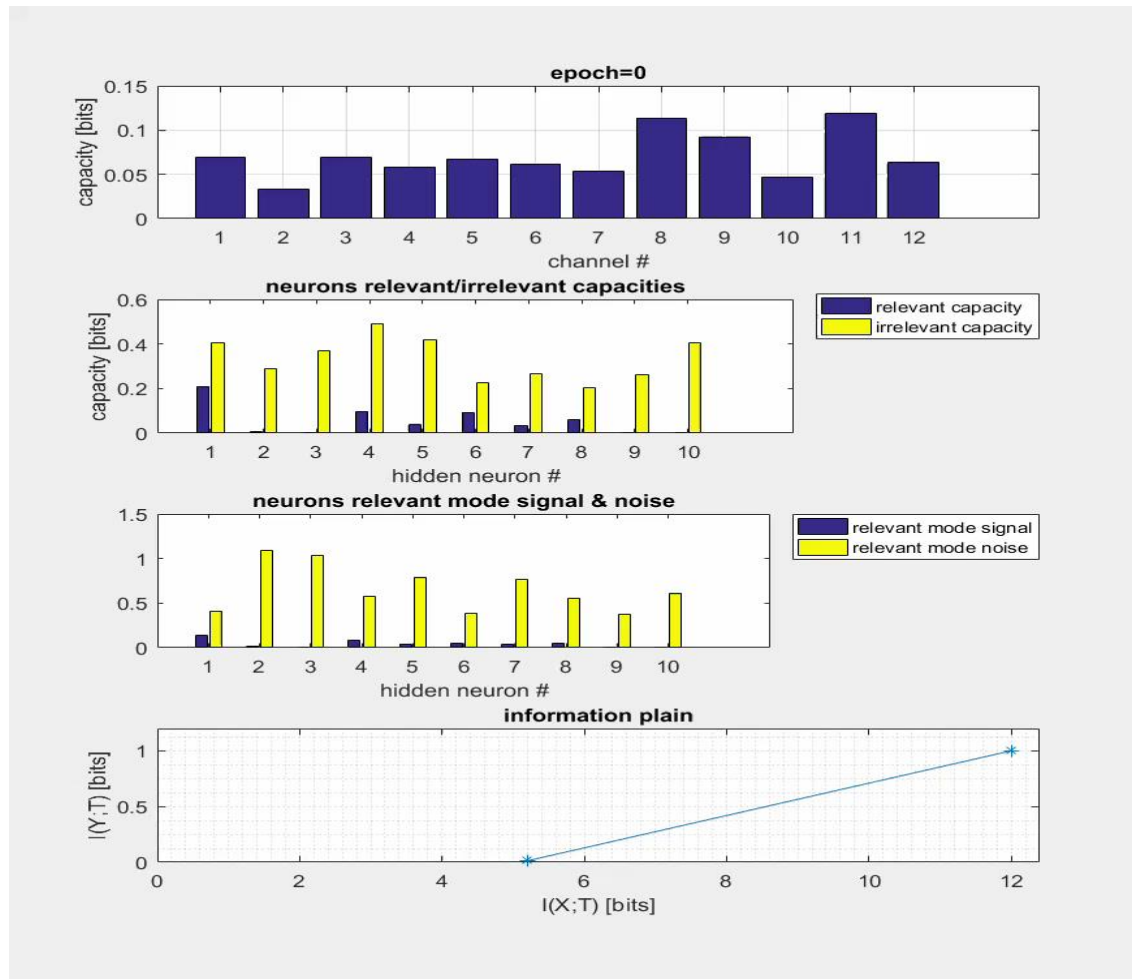
⇒ In optimized DNN only the relevant information $I(X;Y)$ flows through the network

⇒ $SNR_k \approx const.$ we see this in the simulations.

This can determine the final layers locations in the Information Plane...

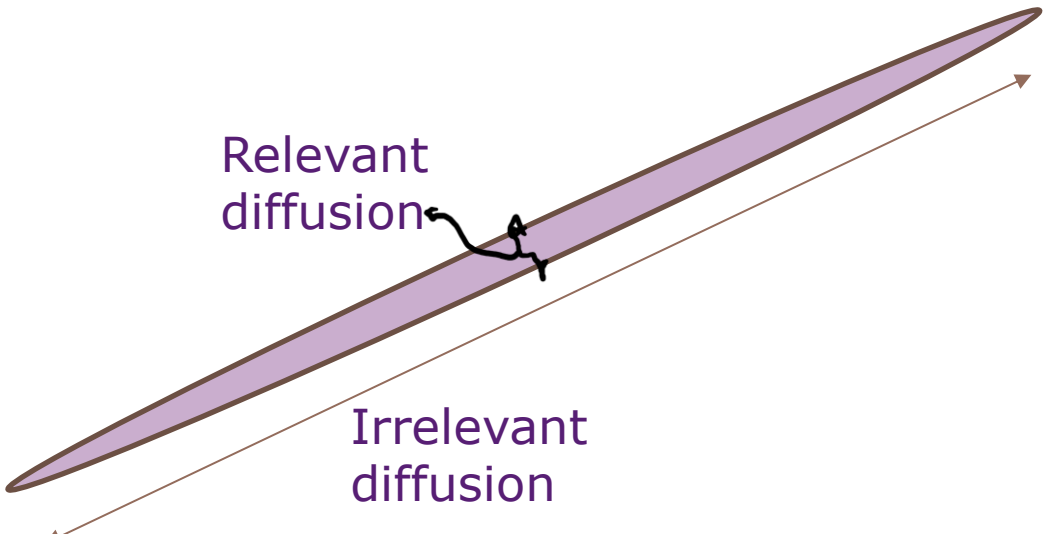
Unless the stochastic relaxation stops through **critical slowing down** near **phase transitions** !

Layers relevant capacity equilibration



Relevant and Irrelevant local dimensions

The covariance matrix of the gradients is very narrow in the relevant local dimensions and very wide in the many other dimensions.

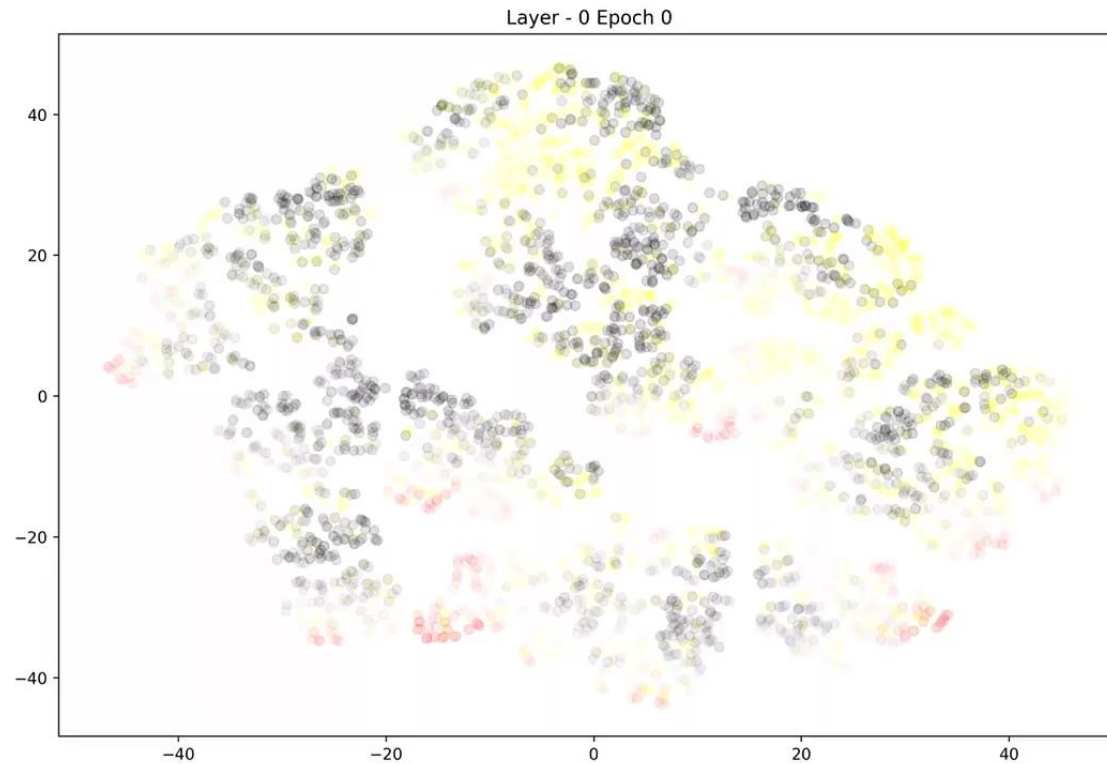
$$\begin{aligned} W^k &\rightarrow W_{cca}^k + \delta W^k \\ T^{k+1} &= \sigma(W_{cca}^k T^k + \xi^k) \\ \xi^k &= \delta W^k T^k \sim N(0, \text{cov}(\delta W^k)) \\ I(T_k; T_{k+1}) &\leq \frac{1}{2} \log \left(1 + \frac{\|W_{cca}\|}{\|\delta W\|} \right) \end{aligned}$$


Relevant diffusion

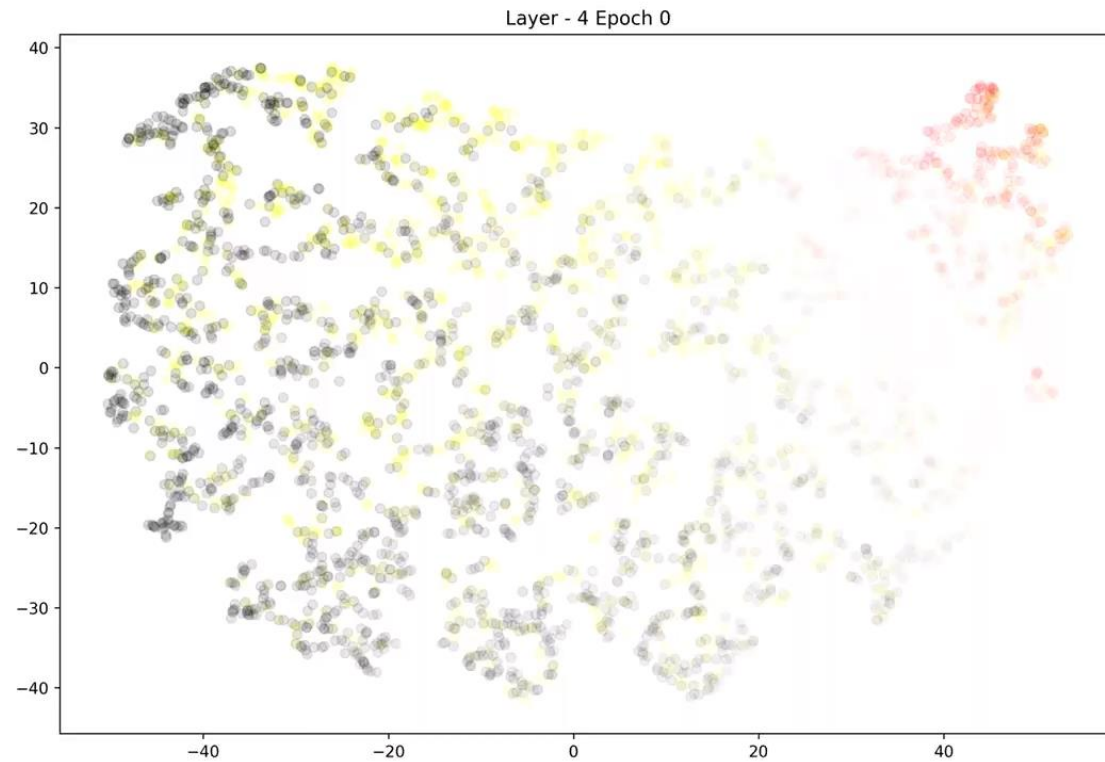
Irrelevant diffusion

$\text{cov}(\delta W^k)$

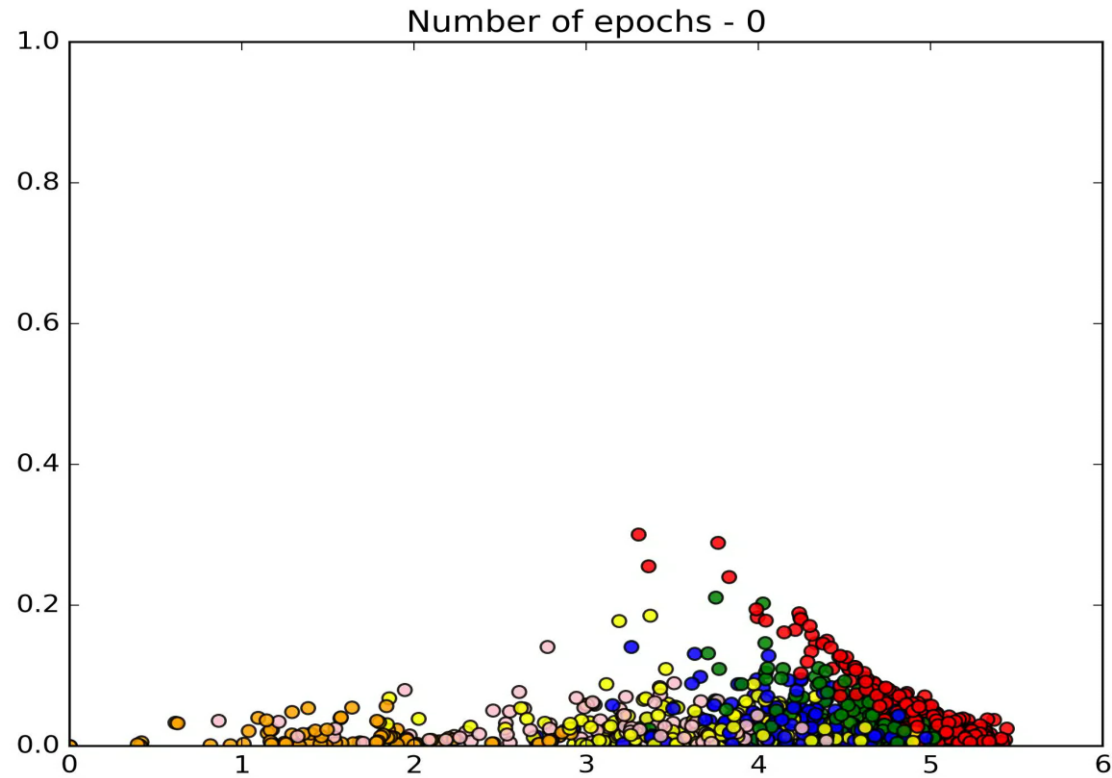
Layers 0 T-SNE Representation



Layer 4 T-SNE Representation



What do single neurons represent?



The IB Bifurcation (phase-transitions) Points



The IB bifurcation points can be found as follows:

$$p_{\beta}(x|\hat{x}) = \frac{p(x)}{Z(x,\beta)} \exp(-\beta D[p(y|x) \| p_{\beta}(y|\hat{x})])$$

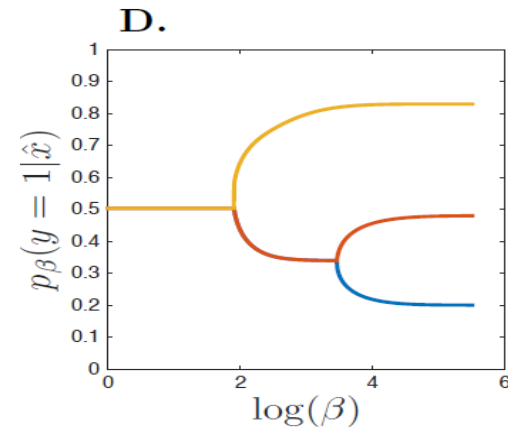
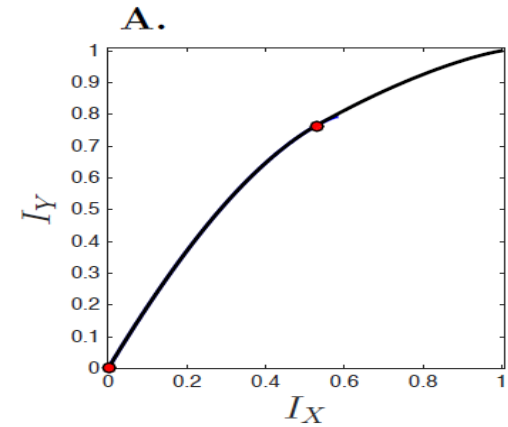
or
$$\ln p_{\beta}(x|\hat{x}) = \ln \frac{p(x)}{Z(x,\beta)} - \beta D[p(y|x) \| p_{\beta}(y|\hat{x})]$$

then:

$$\frac{\partial \ln p_{\beta}(x|\hat{x})}{\partial \hat{x}} = \beta \int_y p(y|x) \frac{\partial \ln p_{\beta}(y|\hat{x})}{\partial \hat{x}}$$

similarly:
$$p_{\beta}(y|\hat{x}) = \int_x p(y|x) p_{\beta}(x|\hat{x})$$

$$\frac{\partial \ln p_{\beta}(y|\hat{x})}{\partial \hat{x}} = \frac{1}{p_{\beta}(y|\hat{x})} \int_x p(y|x) p_{\beta}(x|\hat{x}) \frac{\partial \ln p_{\beta}(x|\hat{x})}{\partial \hat{x}}$$



The IB Bifurcation (phase-transitions) Points



Defining the matrices:

$$\mathbf{C}_{xx'}(\hat{x}, \beta) = \sum_y \frac{p(y|x)}{p_\beta(y|\hat{x})} p_\beta(x'| \hat{x}) p(y|x')$$

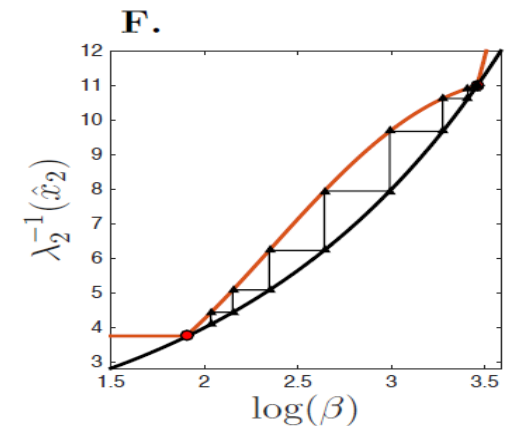
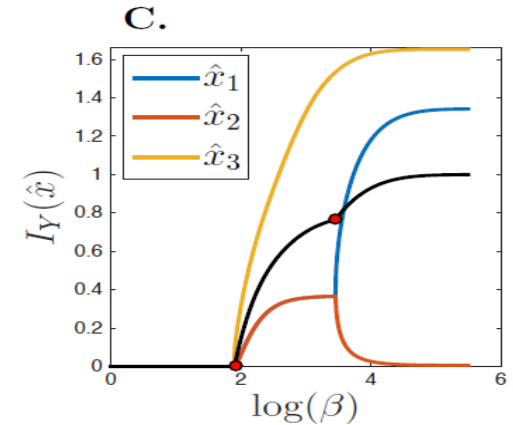
$$\mathbf{C}_{yy'}(\hat{x}, \beta) = \sum_x \frac{p(y|x)}{p_\beta(y|\hat{x})} p_\beta(x|\hat{x}) p(y'|x)$$

these equations can be combined into two (non-linear) eigenvalue problems:

$$\left[I - \beta \mathbf{C}_{xx'}(\hat{x}, \beta) \right] \frac{\partial \ln p_\beta(x'|\hat{x})}{\partial \hat{x}} = 0$$

$$\left[I - \beta \mathbf{C}_{yy'}(\hat{x}, \beta) \right] \frac{\partial \ln p_\beta(y'|\hat{x})}{\partial \hat{x}} = 0$$

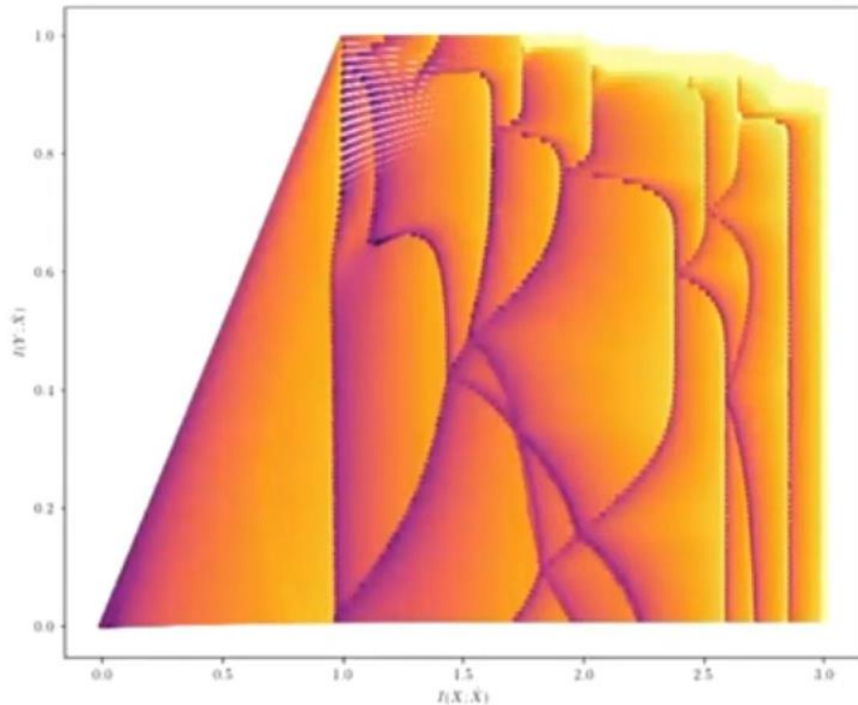
These eigenvalue problems have non-trivial solutions (eigenvectors) only at the critical bifurcation points (second order phase transitions).



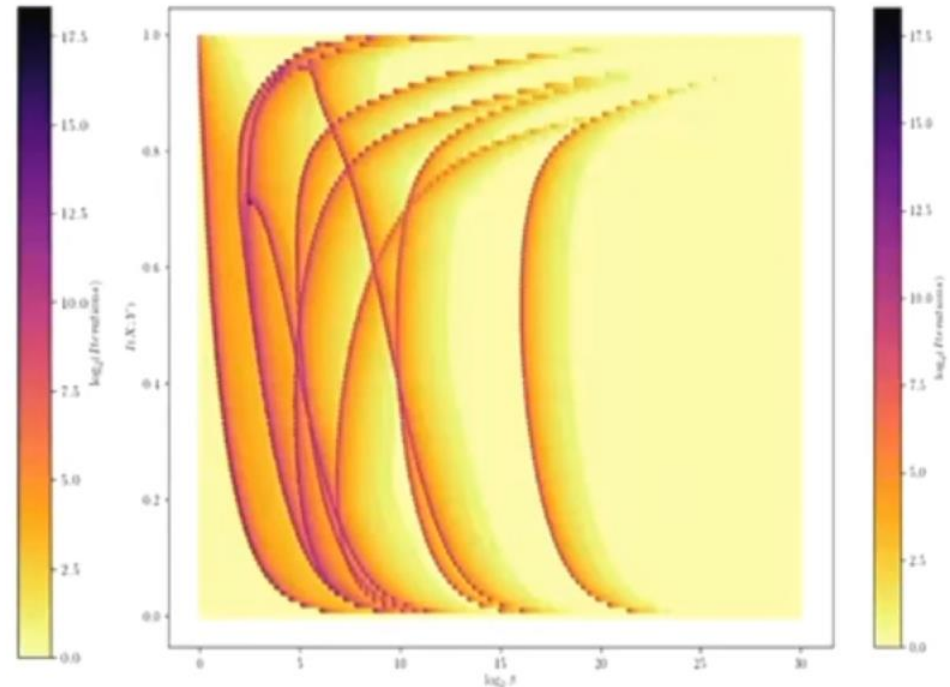
The phase Transition Signature



The deterministic limit



Critical frontiers in the info-plane



Critical frontiers in the
 $I_{xy} - \log \beta$ plane

The Compression Theorem



Theorem: During the diffusion phase, the Mutual Information between consecutive layers decrease with the number of SGD iterations, t , like:

$$I(T_k; T_{k+1}) \leq C_k + t^{-\alpha} R$$

where C_k depends only on the relevant dimension at the k layer and α is the diffusion exponent. R - the “power” of the target, depends only on the problem - not on the architecture.

The Compression Theorem: Proof idea



After decomposition in the singular directions of W_{cca}^k

$$I(T^k; T^{k+1}) \leq \frac{1}{2} \log \left(1 + \frac{\|W_{cca}^k\|}{\|\delta W^k\|} \right) = \frac{1}{2} \sum_i \log \left(1 + \frac{A_{ii}}{\lambda_i(t)} \right) \leq \frac{1}{2} \sum_i \frac{A_{ii}}{\lambda_i(t)} = \frac{1}{t^\alpha} \sum_i \frac{A_{ii}}{\lambda_i(0)}$$

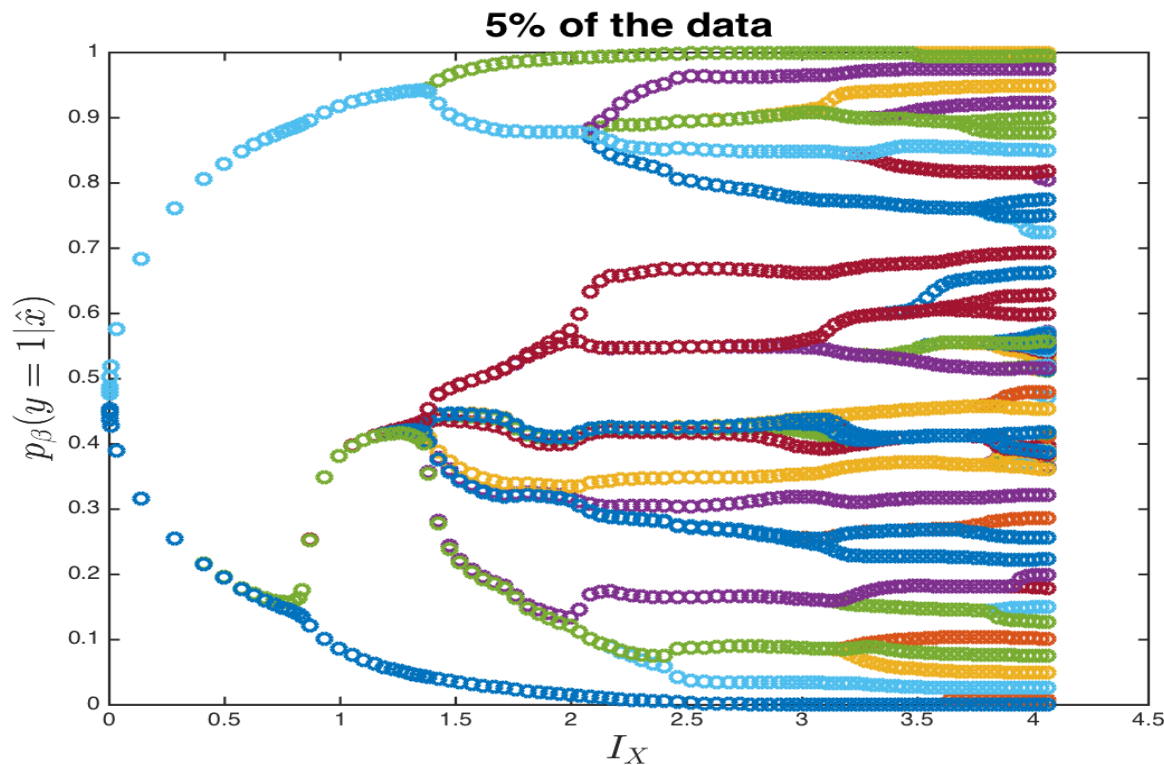
Where the diffusion components grow sub-linearly with time like: $\lambda_i(t) = \lambda_i(0) t^\alpha$

$$I(T_k; T_{k+1}) \leq C_k + t^{-\alpha} R \quad \text{where} \quad R = \sum_i \frac{A_{ii}}{\lambda_i(0)} \quad \text{depends only on the problem!}$$


 If the power R is equally divided between K layers, we get the time power-law scaling : $\Delta t(K) \propto K^{-\frac{1}{\alpha}} \Delta t(1)$

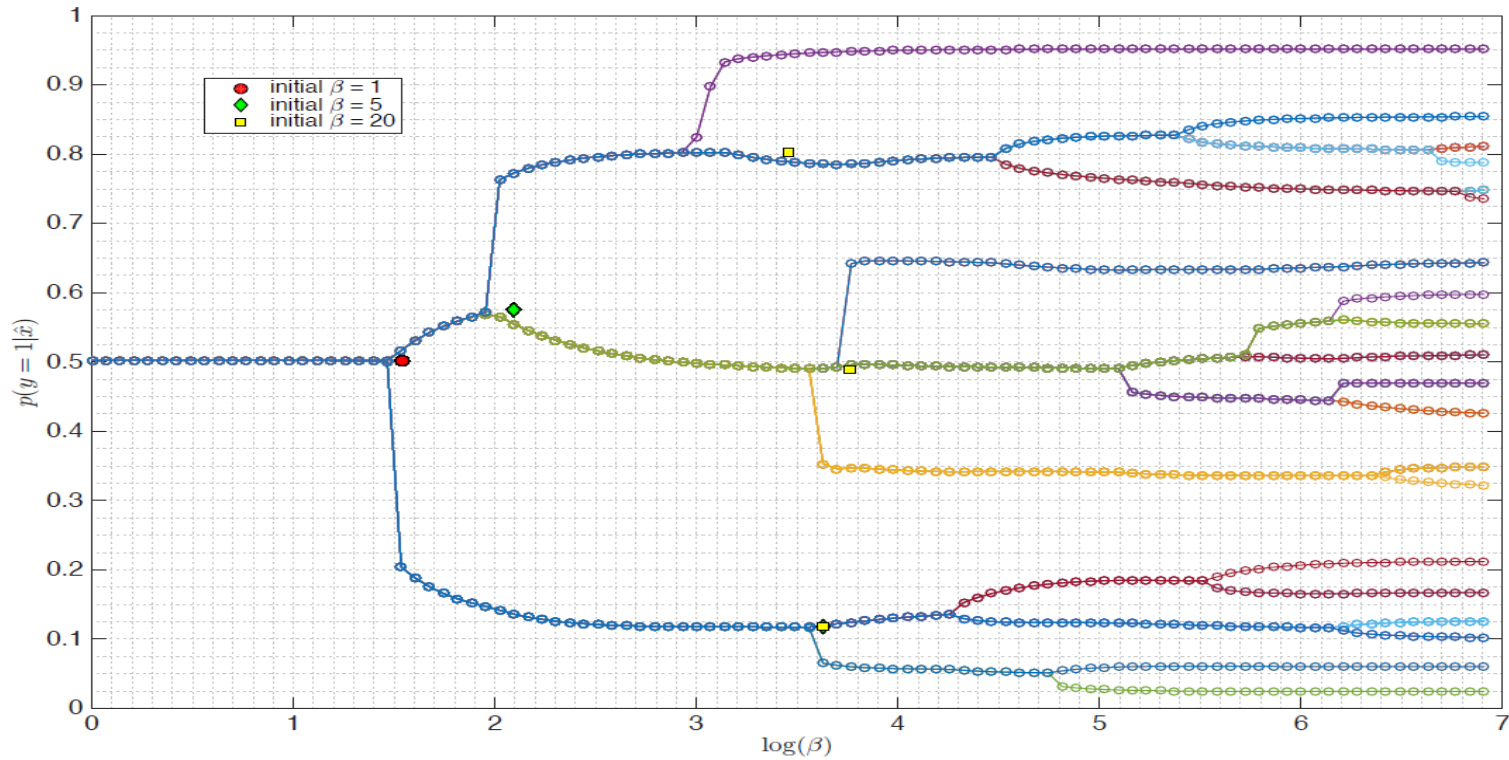
For regular diffusion $0 < \alpha \leq \frac{1}{2}$ and this is at least a quadratic boost in the convergence time !

Bifurcation diagrams in symmetric rule: layers diffusion slows down at phase transitions

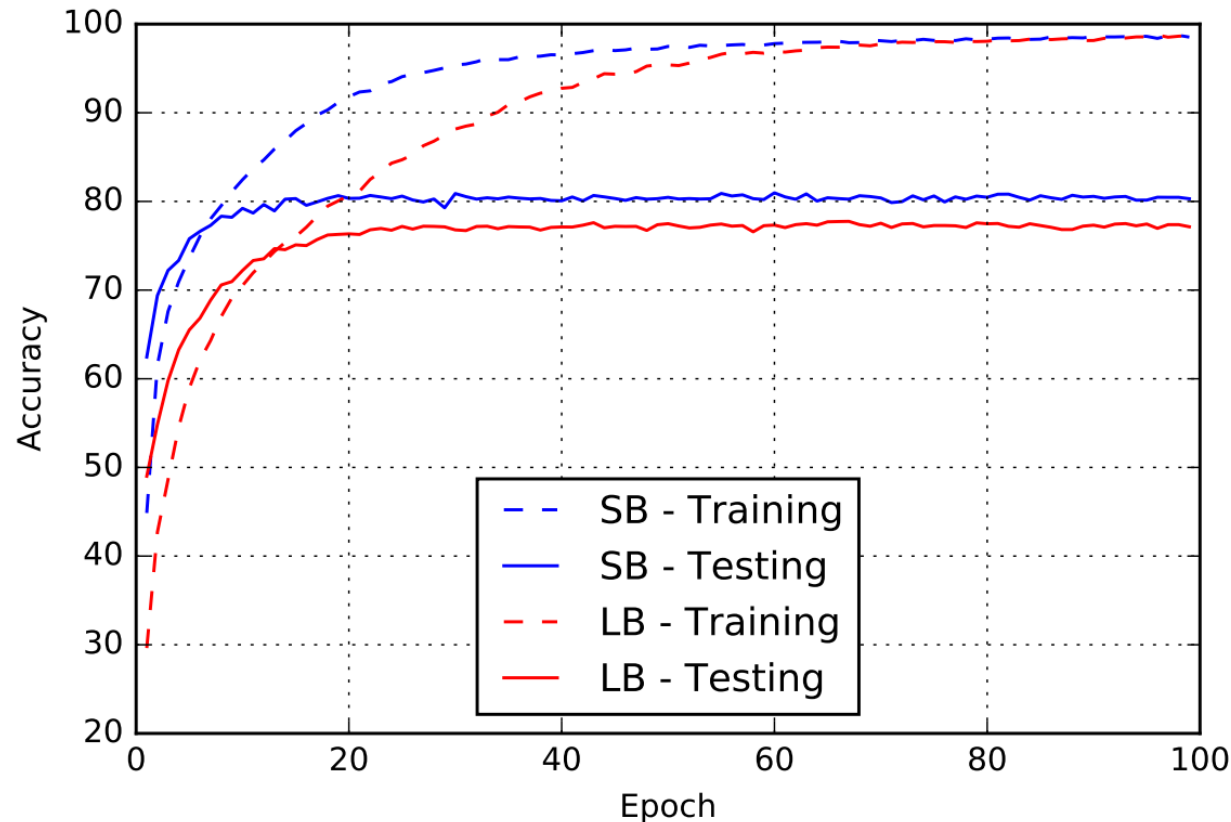


$$W^k \approx \sum_{splits} \frac{\partial \log p(x|t_s^{k-1})}{\partial t_s^{k-1}}$$

Optimal design principles

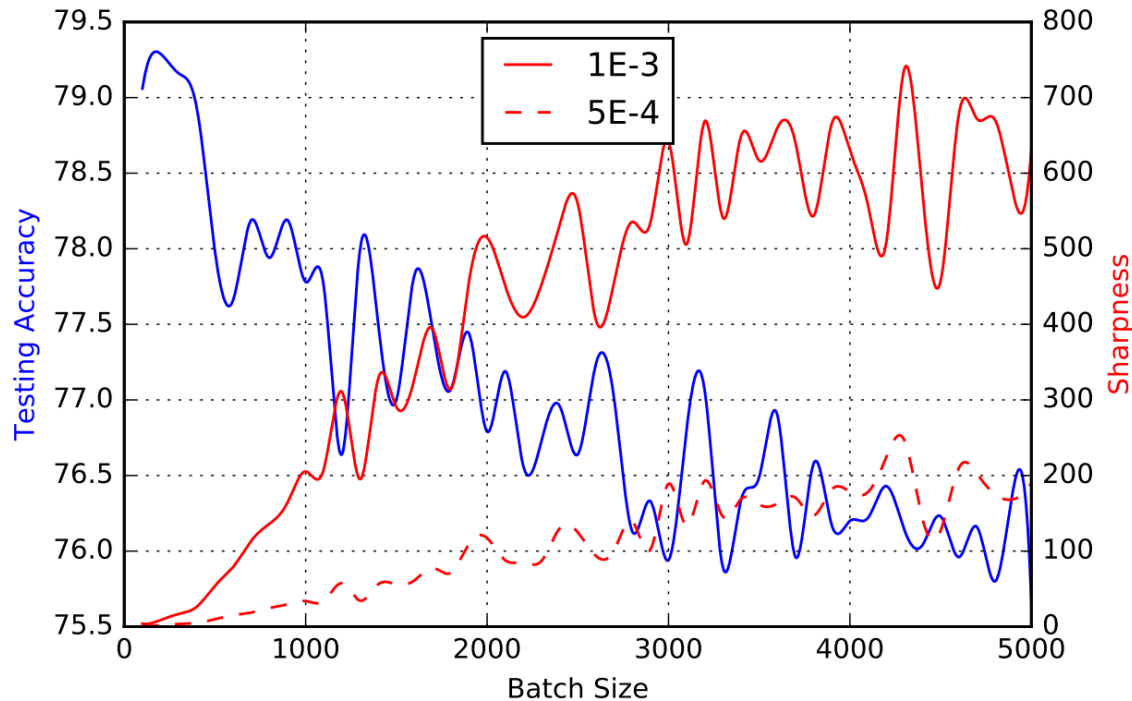


Large batch yields inferior generalization..



Keskar et al: This is due to flatness properties of solutions

Generalization



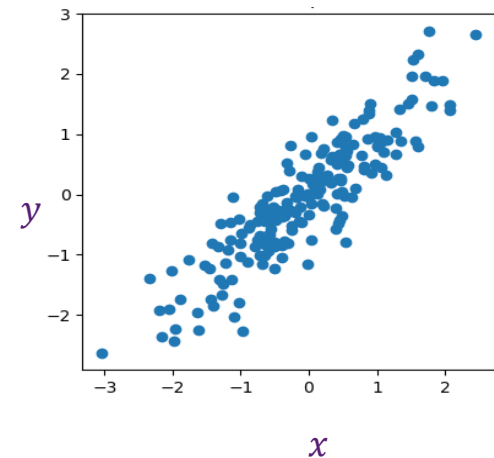
Generalization (negatively) correlates well with sharpness, thus explaining the superiority of small batch over large batch

Estimating Mutual Information through Entropies



Estimate i_{XY} by estimating p_{XY} :

- Binning [**Paninski 03**]
- Kernel [**Moon et al. 95**]
- Nearest neighbor [**Kraskov et al. 04**]



Is there a truly non-parametric estimation?

- Neural estimations [**Poole et al. 18**]



Stochastic dynamics of the training process

- Convergence of SGD to a gibbs (max entropy)
- The compression of information during the learning.
 - Input Compression Bounds
 - PAC-like Generalization Bounds
 - What is a typical input
- The benefits of hidden layers
 - the role of diffusion of the noise at every level
- Many local optimums are good enough..
 - Global optimum is not needed
 - The convergence go through many stages..